Person B Follow-on

Evidence linkage and sequence accuracy audit

Branch	Base	Work mode
codex/person-b-linkage-audit	latest origin/codex/integration	diagnose, then fix only general rules

Purpose

Improve evidence-backed public-sample coverage and exact step-sequence accuracy. The current public benchmark predicts 29 of 32 cases, holds 3 for manual review, and reports sequence accuracy 0.2188. The corrected scorer reports step precision 0.9509, recall 0.8031, and F1 0.8708. These are public diagnostics, not hidden-set results.

Start from the current integration branch

- Fetch first: `git fetch origin --prune`. Create `codex/person-b-linkage-audit` from fetched `origin/codex/integration`. Do not rewrite the completed evidence-reliability branch.
- Reproduce baseline with `python code/evaluation/run_public_benchmark.py`. Save before metrics and case-level diffs in your branch notes.
- Use participant-facing dataset files and public sample output only. Never inspect, import, or infer organizer-only answers.

Investigation

Build a case-by-case evidence trace for sequence mismatches and the 3 manual-review cases. Follow resolved current and historical identifiers, evidence content, resource links, timestamps, policy category, and audit cutoff. Explain each missing or extra step with the participant-visible evidence and rule that supports it.

Question	What to establish
Identity link	Which current or historical identifier links the person to this evidence? Cite file, activity, or audit IDs.
Time correctness	Was evidence captured and observed by audit_at? Which timestamp controls when nested timestamps conflict?
Resource state	Does evidence prove present, absent, redacted, anonymized, active, or blocked state, or is it only an operational claim?
Policy and steps	Which supplied rule determines treatment? Are findings, remediation, verification, citations, and ordering supported?

Implementation and handoff

Fix a general evidence or reconstruction rule only when the trace supports it. A benchmark label by itself is not evidence.

Implementation scope

- Prioritize repeatable issues in identity resolution, cross-file resource linkage, timestamp precedence, state reconstruction, or policy-aware step generation.
- Do not add case-ID rules, sequential resource-ID guesses, hardcoded labels, synthetic shortcuts, or fabricated evidence citations.
- For each retained behavior change, add a small fictional regression test that fails before the fix and passes after it. Cover audit cutoffs and conflicting or missing evidence where relevant.
- Do not modify the UI, package layout, evaluation scorer, task PDF, dataset, or output generation unless diagnosis proves a needed change in your assigned engine scope.

Acceptance criteria

- A written trace explains all investigated missing and extra steps and the 3 manual-review cases, with evidence IDs and cutoff reasoning. Unsupported cases remain unresolved.
- Any code change is general, deterministic, policy-compliant, and covered by regression tests. No participant PII is copied into reports or logs.
- Run relevant tests and `python code/evaluation/run_public_benchmark.py`. Report before/after status accuracy, step precision/recall/F1, sequence accuracy, and coverage.
- Public metrics must not regress without an explicit, evidence-backed explanation. If no safe fix improves them, submit the diagnosis without speculative code changes.
- Commit and push only to `codex/person-b-linkage-audit`. Include findings, changed files, test commands/results, and limitations.

Handoff checklist

Deliver	Branch commit, evidence trace, fictional regression tests, benchmark before/after.
Do not deliver	Case-specific predictions, guessed IDs, organizer-only data, secrets, unsupported citations.

Why this task matters

Exact sequence scoring penalizes missing, extra, incorrectly cited, or misordered steps. Better linkage can recover legitimate evidence-backed steps; better temporal reasoning can reject stale records. The goal is a more complete and correct audit, not a higher score achieved by guessing.